Experiment No. 11

Title:

Study of Linux Shells and Implementation of a Basic Bash Script Using Variables.

Aim:

To study different types of shells available in Linux, understand their features and characteristics, and to write and execute a simple Bash shell script using variables.

Theory:

Shell in Linux

A shell is a command-line interpreter that processes user commands and communicates with the kernel. It allows users to execute commands, manage files, and write scripts for automation.

Types of Shells

Common shells available in Linux include:

Shell	Description
sh	Original Bourne Shell
bash	Bourne Again Shell
ksh	Korn Shell
csh	C Shell
zsh	Z Shell

Bash Shell

Bash is an enhanced version of sh that supports:

- Command history
- Job control

- Scripting with variables and control structures

Shell Scripts

A shell script is a file containing a sequence of commands executed by the shell. It typically starts with a shebang (#!/bin/bash) that specifies the interpreter.

Commands Used

Commands	Description
cat /etc/shells	List available shells
echo $SHELL	Display current shell
bash	Start Bash shell
sh	Start Bourne shell
chmod	Change file permissions
./script.sh	Execute shell script

Procedure / Steps

A. Identifying Different Shells:

Step 1: Login

- Log in to the Linux virtual machine and open the **Terminal**.

Step 2: List available shells using:

- cat /etc/shells

```
┌──(root㉿kali)-[/home/aman_singhal]
└─# cat /etc/shells
# /etc/shells: valid login shells
/bin/sh
/usr/bin/sh
/bin/bash
/usr/bin/bash
/bin/rbash
/usr/bin/rbash
/usr/bin/dash
/usr/bin/tmux
/bin/zsh
/usr/bin/zsh
/usr/bin/pwsh
/opt/microsoft/powershell/7/pwsh
/usr/bin/screen

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

Step 3: Check the current shell using:

- `echo $SHELL`

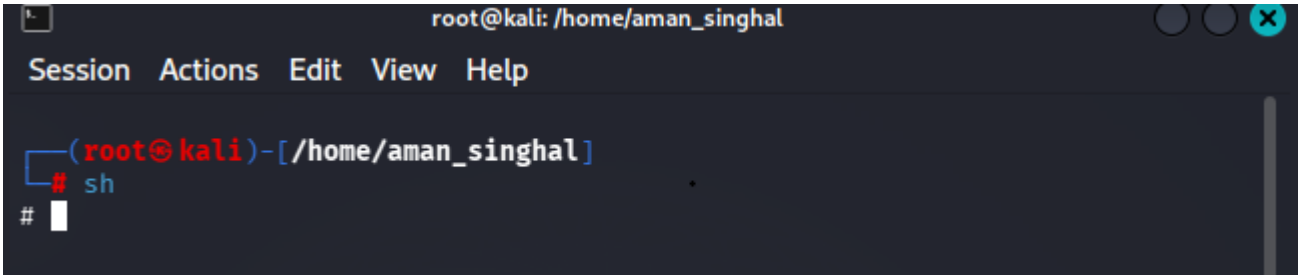
```
┌──(root㉿kali)-[/home/aman_singhal]
└─# echo $SHELL
/usr/bin/zsh

┌──(root㉿kali)-[/home/aman_singhal]
└─#
```

B. Switching Between Shells:

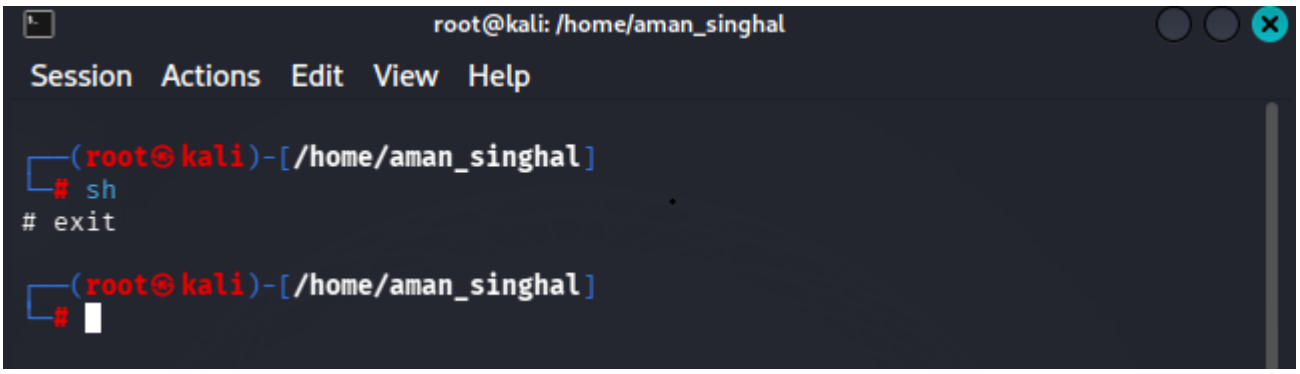
Step 4: Switch to Bourne shell using:

- sh



Step 5: Return to Bash Using:

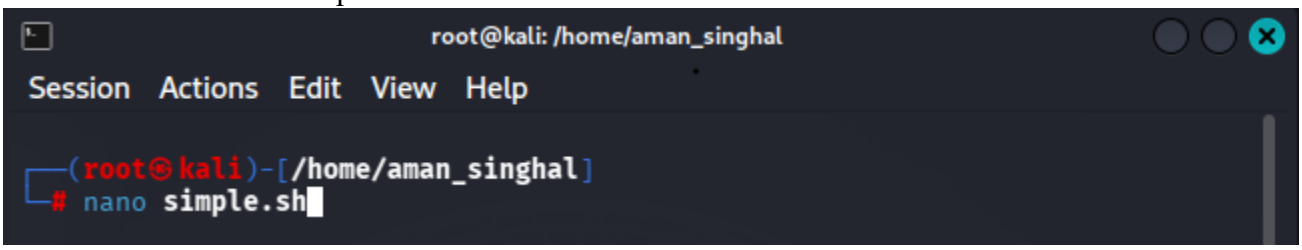
- exit



C. Writing a Simple Bash Script:

Step 6: Create a new script file Using:

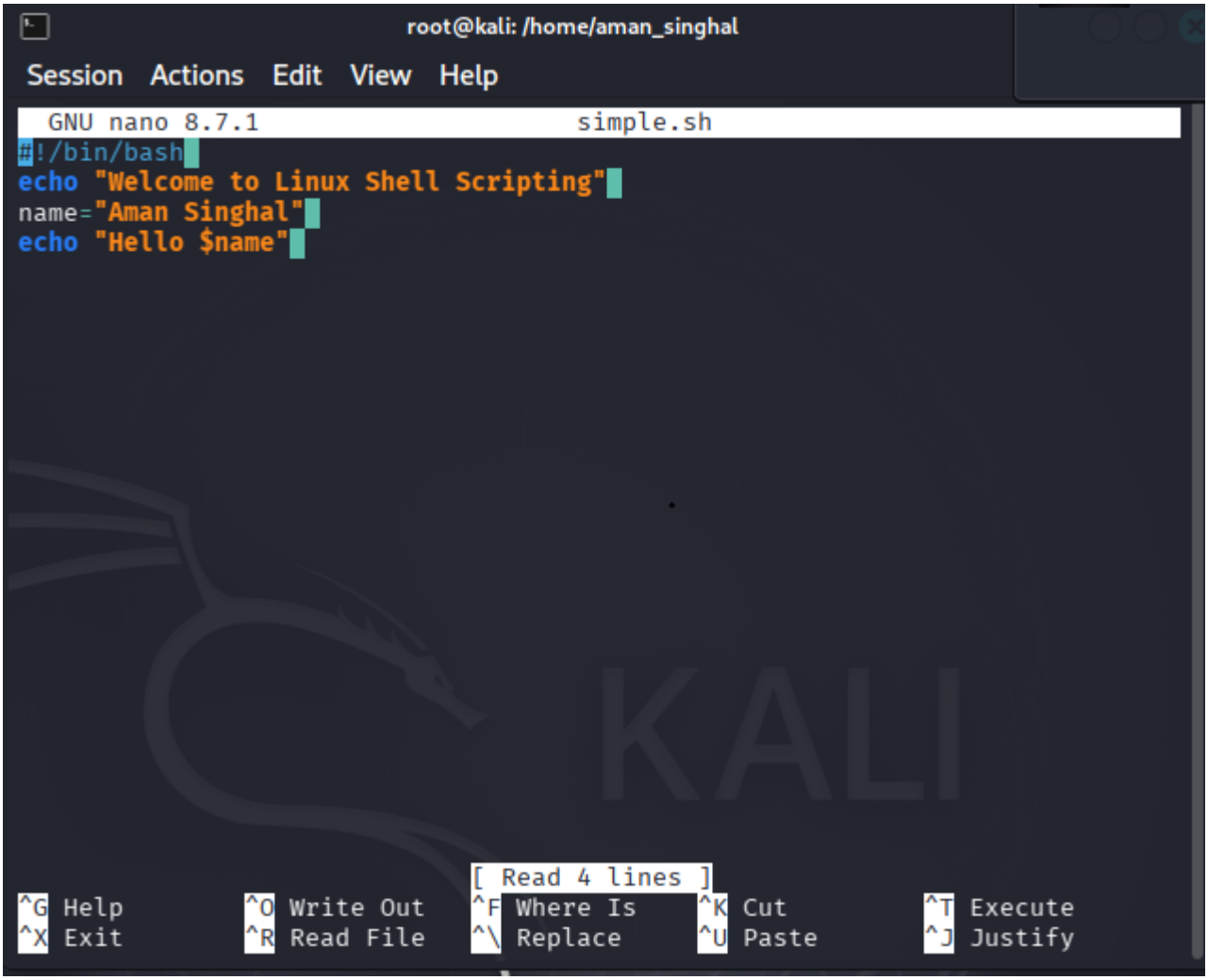
- nano simple.sh



Enter the following script:

- #!/bin/bash
- echo "Welcome to Linux Shell Scripting"
- name="Student"
- echo "Hello $name"

Save and exit the editor.



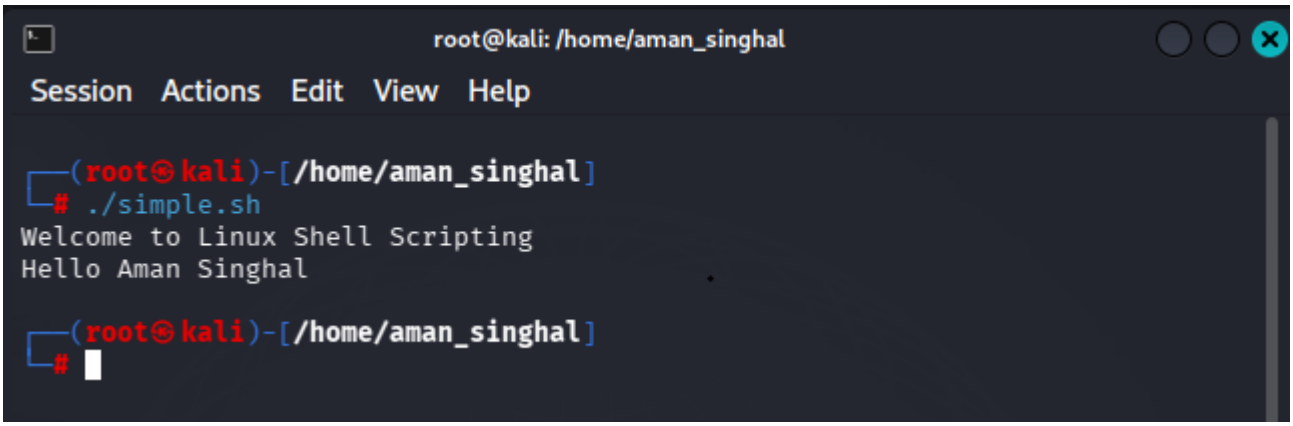
D. Executing the Script:

Step 7: Provide execute permission Using:

- `chmod +x simple.sh`

Step 8: Run the script Using:

- `./simple.sh`



Outcomes:

- Gained understanding of the role of a shell as an interface between user and kernel.
- Learned about various Linux shells and their unique features.
- Acquired practical knowledge of writing and executing Bash scripts.
- Understood the use of variables in shell scripting for storing and manipulating data.
- Developed basic scripting skills useful for automation in Linux environments.

Result:

Different types of Linux shells such as Bash, sh, csh, ksh, and zsh were identified and studied. A simple Bash script was successfully written and executed, demonstrating the use of variables and basic command-line operations.